

A CASE-STUDY BASED REPORT
ON
<Credit Card Fraud Detection System Using Logistic Regression, SVM And Isolation Forest>

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by

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Abstract

Credit card fraud is a major problem in digital financial transactions because fraudulent activities can result in financial losses for customers and financial institutions. As the number of online and card-based transactions increases, identifying fraudulent transactions manually becomes difficult. Machine learning can help analyse transaction information and identify patterns that may indicate fraudulent activity. This case study focuses on automated credit card fraud detection using three machine learning algorithms: Logistic Regression, Support Vector Machine (SVM), and Isolation Forest.

The project uses a credit card transaction dataset and performs exploratory data analysis before applying the machine learning models. The models are evaluated using Accuracy, Precision, Recall, F1-Score, and AUC-ROC. The results show that each algorithm has different strengths in detecting fraudulent transactions. Logistic Regression provides strong Recall, SVM provides a comparable classification performance, and Isolation Forest provides the highest AUC-ROC among the three models. The study demonstrates how machine learning can support automated transaction analysis and help identify potentially suspicious transactions.

Introduction

The rapid growth of digital payments, online banking, and e-commerce has resulted in a large increase in credit card transactions. Along with this growth, the possibility of fraudulent transactions has also increased. Fraudulent transactions may look similar to legitimate transactions, making it difficult to identify them using only manual checking or simple rules. Therefore, automated methods are required to analyse large numbers of transactions efficiently.

Machine learning provides an approach for analysing transaction data and learning patterns associated with legitimate and fraudulent activities. In this project, Logistic Regression, Support Vector Machine, and Isolation Forest are used to analyse credit card transaction data. The performance of these models is compared using different evaluation metrics so that their suitability for fraud detection can be understood.

Background

The project is based on a credit card fraud detection dataset obtained from Kaggle. The project material describes an original dataset containing 20,000 transaction records and 26 attributes. A balanced dataset containing 678 records is also described, consisting of 339 legitimate transactions and 339 fraudulent transactions. The target variable used in the notebook is `is_fraud`.

The dataset contains transaction-related information that can be useful for identifying suspicious activity. The notebook performs several data analysis steps before model implementation. These include checking the dataset shape, columns, data types, statistical summary, missing values, duplicate records, unique values, fraud distribution, numerical and categorical attributes, and possible outliers using the IQR method.

Motivation for the Project

The main motivation for this project is the need for an automated method to identify potentially fraudulent credit card transactions. When a large number of transactions are processed, manually checking every transaction is time-consuming and may not provide consistent results. A machine learning system can analyse transaction information and provide predictions that can support further investigation.

Another motivation is to compare different machine learning techniques for the same fraud detection problem. Logistic Regression and SVM provide supervised learning approaches, while Isolation Forest provides an anomaly-detection approach. Comparing these models using multiple evaluation metrics helps in understanding their different strengths and limitations.

Overview of the Work Carried Out

The project begins with loading the credit card transaction dataset and examining its structure. Exploratory Data Analysis is performed to understand the available attributes and the distribution of fraudulent and legitimate transactions. The analysis includes transaction amount, merchant category, card type, authentication method, transaction channel, device type, foreign transaction status, VPN usage, IP-country mismatch, billing-shipping mismatch, velocity score, merchant risk score, distance from home, transaction count, CVV retry count, prior disputes, customer age, and transaction time.

After the data analysis and preparation, three machine learning algorithms are implemented: Logistic Regression, Support Vector Machine, and Isolation Forest. The models are evaluated using Accuracy, Precision, Recall, F1-Score, and AUC-ROC. Their results are then compared to understand which approach provides useful performance for credit card fraud detection.

Scope and Responsible-Use Statement

The scope of this case study is limited to analysing the supplied credit card transaction dataset and evaluating Logistic Regression, SVM, and Isolation Forest. The purpose is to demonstrate how machine learning can be used for automated transaction analysis and identification of potentially suspicious transactions.

The predictions produced by a machine learning model should be considered as decision-support information rather than final confirmation of fraud. A suspicious transaction may require additional verification or investigation. The results obtained in this project are also limited to the supplied dataset and may change when the models are applied to different transaction data.

Problem Statement

Credit card fraud detection involves identifying fraudulent transactions from a large number of legitimate transactions. Since fraudulent behaviour can vary from one transaction to another, identifying fraud using only fixed rules can be challenging. An effective automated approach should be able to analyse several transaction attributes and identify patterns associated with suspicious activity.

The problem addressed in this case study is to develop and compare machine learning models that can distinguish fraudulent transactions from legitimate transactions. The models should be evaluated using multiple metrics because Accuracy alone may not provide sufficient information about fraud detection performance.

Case-Study Scenario

Consider a financial environment where a large number of credit card transactions are processed every day. Each transaction contains several attributes describing the transaction and related activity. The organisation needs an automated method to analyse this information and identify transactions that may require further attention.

In this case study, the supplied transaction dataset is analysed using exploratory data analysis and three machine learning algorithms. Logistic Regression, SVM, and Isolation Forest are applied to the transaction data, and their results are compared using different evaluation metrics to understand their effectiveness for fraud detection.

Central Case-Study Question

The central question of the study is how effectively Logistic Regression, Support Vector Machine, and Isolation Forest can detect fraudulent credit card transactions using the available transaction attributes.

The study also considers how the three algorithms differ in terms of Accuracy, Precision, Recall, F1-Score, and AUC-ROC. This comparison helps determine the strengths of each model for the fraud detection problem.

Formal Problem Definition

The formal problem is to analyse a dataset containing credit card transaction records and a target variable indicating whether each transaction is fraudulent. The available transaction attributes are used as input information for developing machine learning models that can distinguish fraudulent transactions from legitimate transactions.

The models are evaluated using Accuracy, Precision, Recall, F1-Score, and AUC-ROC. These metrics provide different views of model performance and are important for understanding how well the models identify fraudulent activity and how many incorrect predictions they produce.

Objectives that Guide the Study

The main objective of the study is to analyse the credit card transaction dataset and identify useful patterns related to fraudulent activity. The project also aims to perform exploratory data analysis, prepare the data for machine learning, and implement three different algorithms for fraud detection.

Another objective is to compare Logistic Regression, SVM, and Isolation Forest using multiple evaluation metrics. The comparison helps understand which model provides stronger performance for different fraud detection requirements and demonstrates the usefulness of machine learning in automated transaction analysis.

Assumptions and Limitations

The study assumes that the supplied transaction dataset contains sufficient information to identify patterns related to fraudulent and legitimate transactions. It also assumes that the selected attributes can be used by the machine learning models to generate meaningful predictions. The evaluation is performed using the data available in the project.

The main limitation is that the results are based on the supplied dataset and the three selected algorithms. Model performance may be different when applied to another dataset or a real-world financial environment. The case study therefore demonstrates the machine learning approach within the scope of the available project data rather than representing a complete production fraud detection system.

Dataset and Exploratory Data Analysis

The project uses a credit card fraud dataset obtained from Kaggle. The original dataset contains 20,000 transaction records and 26 attributes. The project material also describes a balanced dataset containing 678 records, with 339 legitimate transactions and 339 fraudulent transactions. The target variable used in the notebook is `is_fraud`, which represents whether a transaction is fraudulent.

Exploratory Data Analysis is performed to understand the structure and characteristics of the dataset before applying machine learning algorithms. The notebook checks the dataset shape, column names, data types, statistical summary, missing values, duplicate records, unique values, and fraud distribution. It also identifies numerical and categorical variables and performs outlier analysis using the IQR method. The analysis considers attributes such as transaction amount, merchant category, card type, authentication method, transaction channel, device type, foreign transaction, VPN usage, IP-country mismatch, billing-shipping mismatch, velocity score, merchant risk score, distance from home, transaction count, CVV retry count, prior disputes, customer age, and transaction time.

Correlation and graphical analysis are also used to understand relationships among the variables.

Algorithm Selection and Implementation

Logistic Regression is selected as one of the supervised machine learning algorithms for the fraud detection problem. It is used to classify transactions based on the available transaction attributes. According to the project results, Logistic Regression achieved an Accuracy of 81.17%, Precision of 10.62%, Recall of 86.83%, F1-Score of 18.07%, and AUC-ROC of 92.40%. The high Recall indicates that the model identifies a large proportion of the fraudulent transactions in the evaluated data.

Support Vector Machine is another supervised learning algorithm used in the project. SVM is evaluated using the same performance measures so that its results can be compared with Logistic Regression. The SVM model achieved an Accuracy of 80.38%, Precision of 11.05%, Recall of 83.25%, F1-Score of 19.41%, and AUC-ROC of 92.63%. These results show that SVM provides slightly higher Precision, F1-Score, and AUC-ROC than Logistic Regression, while Logistic Regression provides higher Accuracy and Recall.

Isolation Forest is used as an anomaly-detection approach for identifying unusual transaction behaviour. The model achieved an Accuracy of 80.02%, Precision of 34.38%, Recall of 17.75%, F1-Score of 23.20%, and AUC-ROC of 94.12%. Among the

three evaluated models, Isolation Forest achieved the highest Precision, F1-Score, and AUC-ROC, while Logistic Regression achieved the highest Accuracy and Recall. These results demonstrate that different algorithms can provide different strengths, and model selection should depend on the specific requirements of the fraud detection application.

Conclusion

This case study demonstrates the use of machine learning algorithms for automated credit card fraud detection. The project combines exploratory data analysis with machine learning implementation to study transaction information and identify potentially fraudulent activity. Logistic Regression, Support Vector Machine, and Isolation Forest are used to analyse the same problem and their results are compared using several evaluation metrics.

The results show that no single metric is sufficient for selecting a fraud detection model. Logistic Regression provides the highest Accuracy and Recall, SVM provides a comparable overall performance, and Isolation Forest provides the highest Precision, F1-Score, and AUC-ROC among the three models. Overall, the study demonstrates that machine learning can provide useful support for automated transaction analysis and help identify suspicious transactions for further investigation.